

Surname (Cognome)	
Name (Nome)	
POLIMI ID Number	
Signature (Firma)	

Politecnico di Milano, June 22nd, 2018

Course on Advanced Computer Architectures

Prof. D. Sciuto, Prof. C. Silvano

EX1	(5 points)	
EX2	(5 points)	
EX3	(6 points)	
Q1	(5 points)	
Q2	(5 points)	
SIX QUESTIONS	(6 points)	
TOTAL	(32 points)	

EXERCISE 1 – TOMASULO (5 points)

Please consider the program in the table be executed on a CPU with dynamic scheduling based on TOMASULO algorithm with:

- 2 RESERVATION STATIONS (**RS1, RS2**) + 2 LOAD/STORE unit (**LDU1, LDU2**) with latency 4
- 2 RESERVATION STATIONS (**RS3, RS4**) + 2 ALU/BR FUs (**ALU1, ALU2**) with latency 2

1. Please complete the TOMASULO TABLE by assuming all cache HITS

ISTRUZIONE	ISSUE	START EXEC	WRITE RESULT	Hazards Type	RSi	UNIT
lw \$f1, 0(\$r1)						
lw \$f2, 0(\$r1)						
fadd \$f2, \$f2, \$f1						
sw \$f2, 0(\$r1)						
lw \$f3, 4(\$r1)						
lw \$f4, 4(\$r1)						
fadd \$f4, \$f4, \$f3						
sw \$f4, 4(\$r1)						
addi \$r1, \$r1, 8						

2. Express the formula then compute the following metrics:

- CPI = _____
- IPC = _____

EXERCISE 2: CACHE PERFORMANCE EVALUATION (5 points)

Let us consider a computer with a memory hierarchy with L1 and L2 caches (including Instructions and Data). Suppose that in **1000** memory references there are **30** misses in L1 and **20** misses in L2.

1. What are the various miss rates?

Miss Rate $L1$ = _____

Miss Rate $L2$ = _____

Miss Rate $L1 L2$ = _____

Assuming: Hit Time $L1$ = 1 clock cycle;

Hit Time $L2$ = 12 clock cycles;

Miss Penalty $L2$ = 120 clock cycles;

2. Write the formula of the **AVERAGE MEMORY ACCESS TIME (AMAT)** when there are L1 and L2 caches:

AMAT = _____

3. How much is the AMAT in this case?

AMAT = _____

4. Given Memory Accesses Per Instruction = **1.5** and $CPI_{EXEC} = 2$, compute the **CPU time** considering the memory hierarchy and **1000** instructions with $T_{CLK} = 1ns$:

CPUtime = _____

EXERCISE 3: CACHE COHERENCY (6 points)

1. To implement cache coherency one of the protocols is the **MESI concurrency protocol** implemented for a **write-invalidate write-back cache**. Please complete the following table with **Yes/No/Maybe** and briefly explain the protocol.

State	Cache up to date?	Memory up to date?	Others have a copy?	Cache can respond other's reads?
Modified				
Exclusive				
Shared				
Invalid				

1. Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches)

Cycle	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid	Yes	Yes
1	P1: write block 0	Modified (0)	Invalid	No	Yes
2	P2: read block 0				
3	P2: write block 0				
4	P1: read block 1				
5	P2: read block 1				
6	P1: write block 1				
7	P2: write block 1				
8	P1: read block 0				
9	P2: write block 1				
10	P2: read block 1				
11	P1: read block 1				
12	P2: write block 1				

QUESTION 1: VLIW ARCHITECTURES (5 points)

1. Describe the architectures of Very Long Instruction Word processors

2. What are possible issues associated with VLIW architectures?

(For each of the answers motivate **why** it is TRUE or FALSE)

Answer 1: Compilers can detect parallelism only in local portions of the code **T** **F**

Answer 2: Larger code size **T** **F**

Answer 3: Increased hw complexity **T** **F**

Answer 4: Less code portability **T** **F**

An ideal dynamic-scheduled CPU should:

- Explain why real processors cannot implement each one of these points.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SIX QUESTIONS: (6 points)

Answer with **True** or **False** to the following statements, considering that some answer may have multiple True answers.

Q1) Which of the following static branch prediction method is preferred to be used in case of loops in the MIPS processor?

Answer 1: Branch always taken	T	F
Answer 2: Branch always not-taken	T	F
Answer 3: Delayed branch with delay slot scheduled from target	T	F
Answer 4: Delayed branch with delay slot scheduled from fall-through	T	F
Answer 5: Backward Taken Forward Not Taken	T	F

Q2) Consider the following Assembly code:

```
      SUB R3, R6, R7
LOOP: BEQ R1, R2, DONE
      ADD R5, R5, R4
      ADDI R1, R1, 1
      J    LOOP
DONE: ...
```

Knowing that: R2 is initialized at 100 & R1 is initialized at 0
How many BHT accesses are performed?

Answer 1: 101	T	F
Answer 2: 200	T	F
Answer 3: 402	T	F
Answer 4: 302	T	F

Q3) Which of the following cache improvements are effective in reducing the miss penalty?

Answer 1: Adopting higher associativity in designing the cache to reducing the impact of collisions when the correct block must be loaded in the cache.	T	F
Answer 2: Avoiding address translation during indexing of the cache to minimize the management time of miss penalties.	T	F
Answer 3: Combining a small and fast level-1 cache with a slower but larger level-2 cache to capture many accesses that would go to the main memory.	T	F
Answer 4: Giving priority to read misses over writes, exploiting techniques to serve the reads before the writes have been completed.	T	F

Q4) Which of the following statements are true in a SIMD architecture ?

Answer 1: SIMD architecture can exploit significant data parallelism.	T	F
Answer 2: Each instruction can operate on multiple loaded data in a single operation.	T	F
Answer 3: Multiple program counters allow multiple instructions execution.	T	F
Answer 4: Given N PEs (Processing Elements), a 2^N Array controller is required to handle them	T	F

Q5) *Considering hardware parallelism techniques, we can say that Vector Architectures and Graphic Processor Units:*

Answer 1: Exploit data-level parallelism by applying a single instruction to a collection of data in parallel. **T F**

Answer 2: Exploits either data-level parallelism or task-level parallelism in a tightly coupled HW model that allows interaction among threads **T F**

Answer 3: Exploits parallelism among largely decoupled tasks specified by the programmer or the OS **T F**

Answer 4: Exploits data-level parallelism at modest level through compiler techniques such as pipelining and at medium levels using speculation **T F**

Q6) *In vector architectures, is it correct to say that: “With Vector Chaining, the system must wait for the last element of result to be written before starting dependent instruction”? (answer True or False)*

Answer: **T F**